Design and Simulation of a Motorized Ball Screw Actuator for a Smart Rehabilitation Knee Exoskeleton

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1 Introduction and Problem Statement

Traditional robotic knee exoskeletons often use rigid actuation that can't adapt to the changing biomechanical needs of a user. This lack of variable stiffness leads to unnatural gait and poor shock absorption. My focus for this project was to design a linear actuation system (Concept 3) that could transform a passive brace into an active-assist platform.

2 Biomechanical Loading Parameters

To make sure the frame wouldn't fail, we first had to define the human-machine interface. We looked at three critical loading extremes: Weight Acceptance, Heel-off, and the Sit-to-stand transition.

2.1 Ground Reaction Forces (GRF)

At the moment of heel strike, the leg acts as a shock absorber[cite: 355]. We calculated the vertical ground reaction force (F_{GRF}) for a 75kg user with a dynamic impact multiplier of 1.5:

$$F_{GRF} = M \times g \times C = 1104 \text{ N} \tag{1}$$

The peak torque demand was identified as 34Nm right after the heel strikes the ground.

3 Concept 3: Mechanical Architecture

My design philosophy for Concept 3 was to maximize torque density and precision using a motorized ball screw architecture.

3.1 System Components

- **Prime Mover:** A high-torque NEMA 34 Stepper motor was selected to provide static support during rehab cycles.

- **Transmission:** An SFU1204 Ball Screw converts rotation into high-force linear thrust with 90% efficiency.
- **Frame Material:** Aluminum 6061-T6 was chosen for its high strength-to-weight ratio.

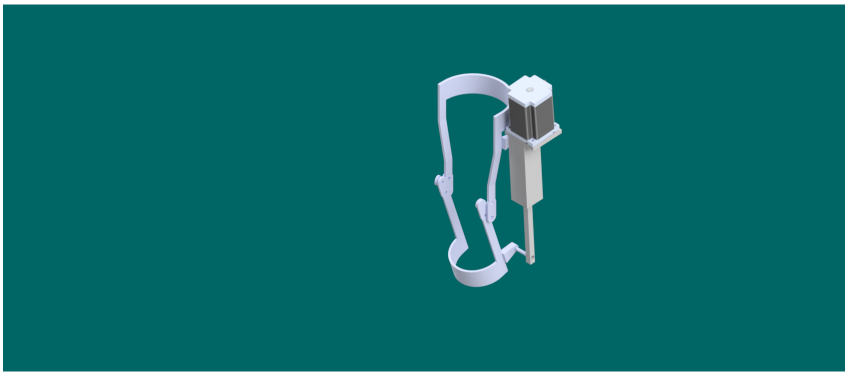

Figure 1: Motorized Ball Screw Actuation System Assembly

4 Calculations and Resolution Analysis

I calculated the theoretical peak stall force for the actuator at 6,361 N, though we set our design threshold at 5,040 N for safety. For a 1,500 N load, the required motor torque is only 1.06 Nm, meaning the NEMA 34 has a massive torque reserve for sit-to-stand impulses.

5 Finite Element Analysis (FEA)

I ran structural simulations in Autodesk to verify the safety factor under the maximum actuator stall condition (5,040 N).

- **Max Stress:** 136.53 MPa.
- **Safety Factor:** 2.01, ensuring no plastic deformation.
- **Max Displacement:** Less than 1.5 mm, which is critical to keep the ball screw aligned so it doesn't bind.

6 Conclusion

The motorized ball screw system provides a rigid and predictable support structure. By combining clinical gait data with inverse dynamics, I was able to verify that this hardware baseline will support a user's weight without stalling.